Adaptive Experimental Design: Posterior CRB, Ignorance Gap, and Adaptive Policies

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Abstract

We develop a self-contained theoretical framework for adaptive experimental design. Starting from the general posterior Cramér–Rao bound (PCRB) and the Tichavský–Muravchik–Nehorai recursion for arbitrary discrete-time state-space models, we successively specialize to a static state, additive Gaussian measurements, and the resulting Bayesian D-optimal problem for *pre-computed* sensor sequences. We then introduce the *ignorance gap* $IG(\boldsymbol{x})$—the log-determinant deficit of the best state-agnostic design relative to an oracle that sees the state—and show that its prior-expectation upper-bounds the improvement achievable by any adaptive policy. When $\mathbb{E}[IG]$ is negligible, the pre-computed design is essentially optimal and no adaptivity is warranted. When $\mathbb{E}[IG]$ is substantial, we define an adaptive policy precisely, contrast it with the pre-computed setting, and characterize the optimal policy via the Bellman equation for the value-to-go W_k. We extend this to the joint optimization of device geometry and adaptive policy using the envelope theorem, prescribe a policy-iteration scheme, and catalog problem classes that exhibit large ignorance gaps. An appendix derives the Bellman equation from first principles.

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1 General PCRB: The Tichavský Recursion

1.1 State-space setting

Consider a discrete-time state-space model:

$$\boldsymbol{x}_k = \boldsymbol{g}_k(\boldsymbol{x}_{k-1}, \boldsymbol{w}_k), \qquad \boldsymbol{w}_k \sim p_{\boldsymbol{w}_k}, \tag{1}$$

$$\boldsymbol{y}_k = \boldsymbol{h}_k(\boldsymbol{x}_k, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim p_{\boldsymbol{v}_k}, \tag{2}$$

where $\boldsymbol{x}_k \in \mathbb{R}^{d_x}$ is the state, $\boldsymbol{y}_k \in \mathbb{R}^{d_y}$ is the observation, $\boldsymbol{s}_k \in \mathbb{R}^{d_s}$ is a controllable sensor parameter at step k, $\boldsymbol{c} \in \mathbb{R}^{d_c}$ collects device-geometry (structural) parameters held fixed across the horizon, $\boldsymbol{w}_k$ is process noise, and $\boldsymbol{v}_k$ is measurement noise. The prior on the initial state is $\boldsymbol{x}_0 \sim p_0$. The sensor parameter $\boldsymbol{s}_k$ may be fixed at design time or chosen adaptively based on $\boldsymbol{y}_{1:k-1}$.

1.2 The Bayesian Information Matrix

Define the *Bayesian Information Matrix* (BIM) at step k as:

$$\boldsymbol{J}_k \;=\; \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_k}\nabla_{\boldsymbol{x}_k}^{\top}\ln p(\boldsymbol{x}_k \mid \boldsymbol{y}_{1:k})\Big], \tag{3}$$

where the expectation is over the joint distribution $p(\boldsymbol{x}_{0:k}, \boldsymbol{y}_{1:k})$. For any (conditionally or asymptotically) unbiased estimator $\hat{\boldsymbol{x}}_k$ of $\boldsymbol{x}_k$ given $\boldsymbol{y}_{1:k}$, the PCRB is:

$$\mathbb{E}\Big[(\hat{\boldsymbol{x}}_k - \boldsymbol{x}_k)(\hat{\boldsymbol{x}}_k - \boldsymbol{x}_k)^{\top}\Big] \;\succeq\; \boldsymbol{J}_k^{-1}. \tag{4}$$

1.3 The Tichavský recursion

Theorem 1 (Tichavský, Muravchik & Nehorai, 1998)**.** *Under regularity conditions (twice-differentiable log-densities, finite Fisher information), the BIM satisfies the recursion*

$$\boldsymbol{J}_{k+1} = \boldsymbol{D}_k^{22} - \boldsymbol{D}_k^{21}\left(\boldsymbol{J}_k + \boldsymbol{D}_k^{11}\right)^{-1}\boldsymbol{D}_k^{12} + \boldsymbol{H}_{k+1}, \tag{5}$$

where the $d_x \times d_x$ matrices arise from the state transition and the measurement:

$$\boldsymbol{D}_k^{11} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_k}\nabla_{\boldsymbol{x}_k}^{\top}\ln p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k)\Big], \tag{6}$$

$$\boldsymbol{D}_k^{12} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_k}\nabla_{\boldsymbol{x}_{k+1}}^{\top}\ln p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k)\Big], \tag{7}$$

$$\boldsymbol{D}_k^{21} = \left(\boldsymbol{D}_k^{12}\right)^{\top}, \tag{8}$$

$$\boldsymbol{D}_k^{22} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_{k+1}}\nabla_{\boldsymbol{x}_{k+1}}^{\top}\ln p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k)\Big], \tag{9}$$

$$\boldsymbol{H}_{k+1} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_{k+1}}\nabla_{\boldsymbol{x}_{k+1}}^{\top}\ln p(\boldsymbol{y}_{k+1} \mid \boldsymbol{x}_{k+1}, \boldsymbol{s}_{k+1})\Big]. \tag{10}$$

The recursion is initialized with the prior information,

$$\boldsymbol{J}_0 = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_0}\nabla_{\boldsymbol{x}_0}^{\top}\ln p_0(\boldsymbol{x}_0)\Big] = \boldsymbol{\Sigma}_0^{-1} \qquad \textit{(if } p_0 \textit{ is Gaussian with covariance } \boldsymbol{\Sigma}_0\textit{).} \tag{11}$$

Remark 1. *The $\boldsymbol{D}_k^{ij}$ matrices depend only on the state dynamics; $\boldsymbol{H}_{k+1}$ is the expected measurement Fisher Information Matrix (FIM) of $p(\boldsymbol{y}_{k+1} \mid \boldsymbol{x}_{k+1}, \boldsymbol{s}_{k+1})$ with respect to $\boldsymbol{x}_{k+1}$. The PCRB* (4) *holds for* any *choice of $\boldsymbol{s}_k$—deterministic or adaptively randomized—provided the expectation in $\boldsymbol{H}_{k+1}$ includes the randomness of the sensor policy.*

2 Specialization: Static State

When the state is static ($\boldsymbol{x}_k = \boldsymbol{x}_0$ for all k, i.e., $\boldsymbol{g}_k$ is the identity with vanishing process noise), the transition becomes deterministic: $p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k) = \delta(\boldsymbol{x}_{k+1} - \boldsymbol{x}_k)$. The $\boldsymbol{D}_k^{ij}$ matrices diverge in this limit, and the Schur complement in (5) collapses.

Limiting argument. Consider additive Gaussian dynamics $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{w}_k$ with $\boldsymbol{w}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_k)$ and take $\boldsymbol{Q}_k \to \boldsymbol{0}$. Direct computation from (6)–(9) gives $\boldsymbol{D}_k^{11} = \boldsymbol{D}_k^{22} = \boldsymbol{Q}_k^{-1}$ and $\boldsymbol{D}_k^{12} = \boldsymbol{D}_k^{21} = -\boldsymbol{Q}_k^{-1}$. The recursion becomes

$$\boldsymbol{J}_{k+1} = \boldsymbol{Q}_k^{-1} - \boldsymbol{Q}_k^{-1}\big(\boldsymbol{J}_k + \boldsymbol{Q}_k^{-1}\big)^{-1}\boldsymbol{Q}_k^{-1} + \boldsymbol{H}_{k+1}. \tag{12}$$

Applying the Woodbury identity $\boldsymbol{Q}^{-1} - \boldsymbol{Q}^{-1}(\boldsymbol{J} + \boldsymbol{Q}^{-1})^{-1}\boldsymbol{Q}^{-1} = (\boldsymbol{Q} + \boldsymbol{J}^{-1})^{-1}$ and taking $\boldsymbol{Q}_k \to \boldsymbol{0}$ gives $(\boldsymbol{Q}_k + \boldsymbol{J}_k^{-1})^{-1} \to \boldsymbol{J}_k$. Hence:

$$\boxed{\boldsymbol{J}_{k+1} = \boldsymbol{J}_k + \boldsymbol{H}_{k+1}, \qquad \boldsymbol{J}_0 = \boldsymbol{\Sigma}_0^{-1}.} \tag{13}$$

Additive information. For a static state, information simply accumulates: each measurement contributes its FIM $\boldsymbol{H}_k$ to the running total. After N steps,

$$\boldsymbol{J}_N \;=\; \boldsymbol{\Sigma}_0^{-1} + \sum_{k=1}^{N} \boldsymbol{H}_k(\boldsymbol{s}_k, \boldsymbol{c}), \tag{14}$$

and the PCRB is

$$\mathbb{E}\Big[(\hat{\boldsymbol{x}} - \boldsymbol{x})(\hat{\boldsymbol{x}} - \boldsymbol{x})^\top\Big] \;\succeq\; \left(\boldsymbol{\Sigma}_0^{-1} + \sum_{k=1}^{N} \boldsymbol{H}_k(\boldsymbol{s}_k, \boldsymbol{c})\right)^{-1}. \tag{15}$$

The additive structure holds whether $\boldsymbol{s}_k$ is fixed or chosen adaptively—adaptivity affects the *value* of each $\boldsymbol{H}_k$, not the sum structure.

3 Specialization: Nonlinear Gaussian Measurement

Now suppose the observation model is

$$\boldsymbol{y}_k = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_k), \tag{16}$$

where $\boldsymbol{h} : \mathbb{R}^{d_x} \times \mathbb{R}^{d_s} \times \mathbb{R}^{d_c} \to \mathbb{R}^{d_y}$ is a possibly nonlinear measurement function and $\boldsymbol{R}_k$ is the noise covariance (possibly depending on $\boldsymbol{s}_k$ and $\boldsymbol{c}$).

Measurement FIM. The log-likelihood is

$$\ln p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k) = -\tfrac{1}{2}(\boldsymbol{y}_k - \boldsymbol{h})^\top \boldsymbol{R}_k^{-1}(\boldsymbol{y}_k - \boldsymbol{h}) + \text{const}, \tag{17}$$

and the measurement Jacobian is $\boldsymbol{F}_k = \partial \boldsymbol{h}/\partial \boldsymbol{x}\big|_{\boldsymbol{x},\boldsymbol{s}_k,\boldsymbol{c}} \in \mathbb{R}^{d_y \times d_x}$. A direct calculation (the cross-term in the Hessian vanishes under $\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}}$) gives the FIM at a specific $\boldsymbol{x}$:

$$\mathrm{FIM}(\boldsymbol{x}; \boldsymbol{s}_k, \boldsymbol{c}) = \boldsymbol{F}_k(\boldsymbol{x})^\top \boldsymbol{R}_k^{-1} \boldsymbol{F}_k(\boldsymbol{x}). \tag{18}$$

Taking the expectation over $\boldsymbol{x}$ (as required by (10)):

$$\boldsymbol{H}_k(\boldsymbol{s}_k, \boldsymbol{c}) = \mathbb{E}_{\boldsymbol{x}}\Big[\boldsymbol{F}_k^\top \boldsymbol{R}_k^{-1} \boldsymbol{F}_k\Big]. \tag{19}$$

For **isotropic** noise $\boldsymbol{R}_k = \sigma^2 \boldsymbol{I}$ this simplifies to $\boldsymbol{H}_k = \sigma^{-2}\,\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{F}_k^\top \boldsymbol{F}_k]$.

Combined static + Gaussian form. Combining Sections 2 and 3, for a static state observed through nonlinear Gaussian measurements with isotropic noise:

$$\boxed{\boldsymbol{J}_N = \boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2} \sum_{k=1}^{N} \mathbb{E}_{\boldsymbol{x}}\Big[\boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})^\top \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})\Big],} \tag{20}$$

with PCRB

$$\mathbb{E}\big[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|^2\big] \;\geq\; \mathrm{tr}\big(\boldsymbol{J}_N^{-1}\big). \tag{21}$$

Remark 2 (Two notions of $\boldsymbol{J}_N$)**.** *The BIM* (20) *places* $\mathbb{E}_{\boldsymbol{x}}[\cdot]$ inside *the sum. For design, we will also use the* state-dependent *information*

$$\boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}) = \boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2} \sum_{k=1}^{N} \boldsymbol{F}_k(\boldsymbol{x})^\top \boldsymbol{F}_k(\boldsymbol{x}), \tag{22}$$

which is a random matrix through $\boldsymbol{x}$*. The two are related by* $\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{J}_N(\boldsymbol{x})] = \boldsymbol{J}_N$*, and by Jensen's inequality (concavity of* $\ln\det$*),*

$$\mathbb{E}_{\boldsymbol{x}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x})] \;\leq\; \ln\det \mathbb{E}_{\boldsymbol{x}}[\boldsymbol{J}_N(\boldsymbol{x})]. \tag{23}$$

We use the same symbol $\boldsymbol{J}_N$ *for both and disambiguate by whether an* $\boldsymbol{x}$ *argument is shown.*

4 Pre-Computed Optimal Sensors: Averaged D-Optimality

4.1 The fixed-design problem

Suppose we commit to a sensor sequence $\boldsymbol{s}_{1:N} = (\boldsymbol{s}_1, \ldots, \boldsymbol{s}_N)$ at design time, *before* any observations are collected. Each $\boldsymbol{s}_k$ is a fixed vector in $\mathbb{R}^{d_s}$, *not* a function of anything. Coupled with the device geometry $\boldsymbol{c}$, this defines a *pre-computed* (or *static*, or *non-adaptive*) design.

Since information accumulates additively under (20) and the PCRB depends on $(\boldsymbol{c}, \boldsymbol{s}_{1:N})$ only through $\boldsymbol{J}_N$, a natural design objective is the Bayesian D-optimal criterion:

$$\boxed{\min_{\boldsymbol{c},\, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_N} \mathbb{E}_{\boldsymbol{x}}[-\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})]\,.} \tag{24}$$

This is a **single-level** optimization over $d_c + Nd_s$ variables. The objective is the expected log-volume of the per-state Cramér–Rao uncertainty ellipsoid ($\propto (\det \boldsymbol{J}_N)^{-1/2}$).

Remark 3 (Why single-level)**.** *The integrand $\sum_k \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})^\top \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})$ is a sum of per-step contributions where each $\boldsymbol{s}_k$ appears in only one summand and the summand does not reference any $\boldsymbol{y}_{1:k-1}$—there is no inner optimization or dependence on observations to unroll. The outer minimization over $(\boldsymbol{c}, \boldsymbol{s}_{1:N})$ is therefore flat (single-level) in $d_c + Nd_s$ variables.*

4.2 Gradient structure

Problem (24) is smooth in $(\boldsymbol{c}, \boldsymbol{s}_{1:N})$ when $\boldsymbol{h}$ is smooth. Componentwise, the integrand's gradient at a fixed $\boldsymbol{x}$ is

$$\frac{\partial}{\partial s_{k,i}}(-\ln\det \boldsymbol{J}_N(\boldsymbol{x})) = -\frac{2}{\sigma^2}\,\mathrm{tr}\left(\boldsymbol{J}_N(\boldsymbol{x})^{-1}\,\boldsymbol{F}_k^\top \frac{\partial \boldsymbol{F}_k}{\partial s_{k,i}}\right), \tag{25}$$

using $\partial \ln\det A/\partial A = A^{-\top}$ (which reduces to A^{-1} on the symmetric PD $\boldsymbol{J}_N$). The expected gradient is obtained by taking $\mathbb{E}_{\boldsymbol{x}}[\cdot]$ (Monte Carlo in practice). The gradient with respect to $\boldsymbol{c}$ is analogous, with $\partial \boldsymbol{F}_k/\partial c_j$ propagated through whatever physics model relates the geometry to the measurement Jacobian.

The $\boldsymbol{J}_N^{-1}$ factor couples all measurements: $\partial/\partial s_{k,i}$ of the objective depends on the information from *all other* measurements through $\boldsymbol{J}_N$. This is structural to $\ln\det$ criteria.

4.3 Comparison of alphabetic criteria

Classical optimal-design theory considers several scalar functions of $\boldsymbol{J}_N$:

Criterion	Objective	Interpretation
A-optimal	$\min \mathrm{tr}(\boldsymbol{J}_N^{-1})$	Minimize average variance
D-optimal	$\min -\ln\det \boldsymbol{J}_N$	Minimize log-volume of confidence ellipsoid
E-optimal	$\min \lambda_{\max}(\boldsymbol{J}_N^{-1})$	Minimize worst-case variance

D-optimality has three properties that make it the natural choice:

- **Reparametrization invariance**: $\det \boldsymbol{J}_N$ transforms covariantly under nonsingular changes of coordinates on $\boldsymbol{x}$ (up to the Jacobian determinant), while $\mathrm{tr}(\boldsymbol{J}_N^{-1})$ does not.
- **Balanced information**: it penalizes near-singular $\boldsymbol{J}_N$ much more severely than A-optimality, forcing the sensor sequence to inform *all* state directions rather than piling effort into the most informative one.
- **Information-theoretic meaning**: for Gaussian posteriors, $\frac{1}{2}\ln\det(2\pi e\,\boldsymbol{J}_N^{-1})$ is the differential entropy, so minimizing $-\ln\det \boldsymbol{J}_N$ minimizes the posterior entropy—equivalently, maximizing the mutual information $I(\boldsymbol{x}; \boldsymbol{y}_{1:N})$.

The criteria are related by AM–GM: $d_x^{-1}\,\mathrm{tr}(\boldsymbol{J}_N^{-1}) \geq (\det \boldsymbol{J}_N^{-1})^{1/d_x}$ with equality when $\boldsymbol{J}_N$ is a scalar multiple of $\boldsymbol{I}$. For small d_x they tend to move together.

5 The Ignorance Gap

The pre-computed design (24) hedges against the unknown $\boldsymbol{x}$ by averaging over the prior. How much worse is this hedging than an oracle that sees $\boldsymbol{x}$ and picks sensors specifically for it? The ignorance gap answers this question.

5.1 Oracle and fixed-design values

Define the *oracle* value at a specific state $\boldsymbol{x}$:

$$V_{\text{oracle}}(\boldsymbol{x}) \;=\; \max_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}). \tag{26}$$

This is the best possible log-information if the true $\boldsymbol{x}$ were known at design time. No realizable policy can exceed it, because no realizable policy has access to $\boldsymbol{x}$.

Define the *fixed-design* value:

$$V_{\text{fixed}} \;=\; \max_{\boldsymbol{s}_{1:N}} \mathbb{E}_{\boldsymbol{x}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})]\,, \tag{27}$$

with optimal pre-computed sensors $\boldsymbol{s}^\star_{1:N}$. This is exactly the (negated) optimal value of (24) for fixed $\boldsymbol{c}$.

5.2 Definition of the ignorance gap

The **ignorance gap at $\boldsymbol{x}$** is

$$\boxed{IG(\boldsymbol{x}) \;=\; V_{\text{oracle}}(\boldsymbol{x}) - \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}^\star_{1:N}, \boldsymbol{c}) \;\geq\; 0.} \tag{28}$$

It measures how much better the sensors *could* be if tailored specifically to this $\boldsymbol{x}$, relative to the one-size-fits-all $\boldsymbol{s}^\star_{1:N}$. By Jensen's inequality and the definition of V_{fixed}, $\mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x})] = \mathbb{E}[V_{\text{oracle}}] - V_{\text{fixed}}$, the prior-expected ignorance cost.

5.3 The expected ignorance gap bounds adaptation

Introduce (a preview of Section 6) the value of the optimal adaptive policy:

$$V_{\text{adaptive}} \;=\; \max_{\pi}\; \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}), \boldsymbol{c})]\,, \tag{29}$$

where the max is over all measurable policies $\pi = \{\pi_k : \boldsymbol{y}_{1:k-1} \mapsto \boldsymbol{s}_k\}$. Since a fixed design is a special (constant) policy, $V_{\text{adaptive}} \geq V_{\text{fixed}}$. Since no policy can beat the oracle, $V_{\text{adaptive}} \leq \mathbb{E}_{\boldsymbol{x}}[V_{\text{oracle}}(\boldsymbol{x})]$. Therefore:

$$\boxed{0 \;\leq\; V_{\text{adaptive}} - V_{\text{fixed}} \;\leq\; \mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x})].} \tag{30}$$

This is the *Expected Value of Perfect Information* (EVPI) bound from decision theory: the expected ignorance gap is an upper bound on the value of the optimal adaptive policy over the best fixed design.

Corollary 1. *If $\mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x})] \approx 0$, then no adaptive policy—no matter how clever—can significantly improve upon the pre-computed design* (24)*.*

Remark 4 (EVPI is necessary but not sufficient for adaptive gain)**.** *The converse of Corollary 1 does* not *hold: a large $\mathbb{E}[IG]$ indicates that adaptation* might *help but does not guarantee it. The inequality $V_{adaptive} - V_{fixed} \leq \mathbb{E}[IG]$ is often quite slack, and the realized gap depends sensitively on the geometry $\boldsymbol{c}$. Two designs with the same $\mathbb{E}[IG]$ can have radically different $V_{adaptive} - V_{fixed}$; a design with smaller $\mathbb{E}[IG]$ can in fact have a* larger *realized adaptive advantage than one with larger $\mathbb{E}[IG]$. The concrete numerical demonstration of this for the radar case study appears in Section 8.5 and is quantified across Gaussian beamwidths in the companion numerics document. Consequently, the joint $(\boldsymbol{c}, \pi)$ optimization of Section 8 cannot be replaced by decoupled "optimize $\boldsymbol{c}$ by EVPI, then run adaptive policy."*

5.4 When is the ignorance gap small?

$IG(\boldsymbol{x}) \equiv 0$ exactly when $\operatorname{argmax}_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$ does not depend on $\boldsymbol{x}$—i.e., the same sensor sequence is optimal regardless of the true state. This holds exactly when $\boldsymbol{F}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is independent of $\boldsymbol{x}$ (linear measurement model).

More generally, $IG(\boldsymbol{x})$ is small when the Jacobian's $\boldsymbol{x}$-dependence is weak relative to its $\boldsymbol{s}$-dependence. A Taylor expansion of a weakly-nonlinear measurement $\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ around the prior mean $\boldsymbol{\mu}_0$ gives $\boldsymbol{F}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) = \boldsymbol{F}_0(\boldsymbol{s}, \boldsymbol{c}) + O(\|\boldsymbol{x} - \boldsymbol{\mu}_0\|)$; if the prior standard deviation is small compared to the scale on which $\boldsymbol{F}$ varies with $\boldsymbol{x}$, the oracle-optimal sensors are nearly $\boldsymbol{x}$-independent and $IG \approx 0$.

5.5 Computability of the ignorance gap

The ignorance gap is directly computable from the pre-computed solution with *no adaptive solver required*:

1. Solve (24) to obtain $(\boldsymbol{c}^\star, \boldsymbol{s}^\star_{1:N})$.
2. For a Monte Carlo sample $\{\boldsymbol{x}^{(i)}\}_{i=1}^M$ drawn from the prior:
   (a) Evaluate $\ln\det \boldsymbol{J}_N(\boldsymbol{x}^{(i)}, \boldsymbol{s}^\star_{1:N}, \boldsymbol{c}^\star)$ (the fixed-design log-det at $\boldsymbol{x}^{(i)}$).
   (b) Solve $\max_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_N(\boldsymbol{x}^{(i)}, \boldsymbol{s}_{1:N}, \boldsymbol{c}^\star)$ (an Nd_s-variable optimization).
   (c) Compute $IG(\boldsymbol{x}^{(i)})$ as their difference.
3. Report sample statistics: $\widehat{\mathbb{E}}[IG]$, $\widehat{\mathrm{std}}[IG]$, $\widehat{\max}[IG]$, $\Pr[IG > \epsilon]$.

If $\widehat{\mathbb{E}}[IG] \ll 1$ nat, Corollary 1 says we are done: the fixed design is essentially optimal. This offline check costs M small optimizations and should precede any investment in an adaptive solver.

Diagnostic statistics. Beyond the mean, the distribution of $IG(\boldsymbol{x})$ is informative:

Statistic	Interpretation
$\mathbb{E}[IG]$	Upper bound on adaptation value (EVPI, in nats)
$\mathrm{std}[IG]/\mathbb{E}[IG]$	Heterogeneity: large $\Rightarrow$ benefit concentrated on certain $\boldsymbol{x}$
$\max IG$	Worst-case suboptimality of fixed design
$\Pr(IG > \epsilon)$	Fraction of state space where adaptation matters

6 Adaptive Policies

When $\mathbb{E}[IG]$ is *not* negligible, the fixed design leaves value on the table: by observing intermediate $\boldsymbol{y}_k$'s we may be able to infer something about $\boldsymbol{x}$ and then use that information to pick later sensors more informatively. This motivates *adaptive* policies.

6.1 Definition of an adaptive policy

An **adaptive policy** is a sequence of measurable functions

$$\pi = \{\pi_1, \pi_2, \ldots, \pi_N\}, \qquad \pi_k : \mathbb{R}^{d_y(k-1)} \to \mathbb{R}^{d_s}, \tag{31}$$

where $\pi_k(\boldsymbol{y}_{1:k-1})$ is the sensor action chosen at step k as a function of *only* the observations collected *before* step k. By convention π_1 takes no argument (nothing has been observed yet), so $\pi_1 \in \mathbb{R}^{d_s}$ is a fixed vector.

Equivalently, since $\boldsymbol{y}_{1:k-1}$ enters only through the posterior $b_k = p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1})$ (the *belief*), we can write $\pi_k(b_k)$—a map from beliefs to sensors.

6.2 Contrast with pre-computed sensors

The difference is operational:

Property	Pre-computed (Section 4)	Adaptive (this section)
$\boldsymbol{s}_k$ is	a fixed vector $\in \mathbb{R}^{d_s}$	a function $\pi_k(\boldsymbol{y}_{1:k-1})$
Chosen	at design time, before any data	at run time, after each observation
Depends on $\boldsymbol{y}_{1:k-1}$?	No	Yes
Search space	$\mathbb{R}^{d_c+Nd_s}$ (finite-dim)	space of functions (infinite-dim)
Information benefit	hedges average-case $\boldsymbol{x}$	can target inferred $\hat{\boldsymbol{x}}$
Upper bound on value	V_{fixed} (27)	$V_{\text{adaptive}} \leq V_{\text{fixed}} + \mathbb{E}[IG]$

The pre-computed setting commits the full sequence $\boldsymbol{s}_{1:N}$ before seeing any $\boldsymbol{y}_k$; the adaptive setting selects $\boldsymbol{s}_k$ after $\boldsymbol{y}_{1:k-1}$ have been revealed. The gap between what each can achieve is upper-bounded by $\mathbb{E}[IG]$.

Remark 5 (Not every run-time computation is "adaptive")**.** *A design that executes sensors $\boldsymbol{s}^\star_{1:N}$ at run time but with* fixed *(pre-computed) values is still a pre-computed design in this sense—even though the sensors are physically deployed during data collection. "Adaptive" refers specifically to whether $\boldsymbol{s}_k$* depends on *$\boldsymbol{y}_{1:k-1}$ as a function, not when the numerical values are computed.*

7 Finding the Optimal Adaptive Policy: Bellman's Equation

Problem (29) is a search over functions, and is at face value intractable. The key simplification is that the posterior b_k is a *sufficient statistic* for the observation history $\boldsymbol{y}_{1:k-1}$—two histories producing the same posterior are equivalent for all future decisions—so we can work with policies on beliefs.

7.1 Value-to-go

Write $\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}) = \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$ for the log-information at state $\boldsymbol{x}$ with sensor sequence $\boldsymbol{s}_{1:N}$. Let $b_k = p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1})$ denote the belief at step k. Define the **value-to-go** $W_k(b_k)$: the best achievable expected log-information from step k onward, starting at belief b_k and using the optimal continuation policy.

7.2 Bellman recursion

The value-to-go satisfies the **Bellman equation**:

$$W_k(b_k) = \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})], \qquad k = N, N-1, \ldots, 1, \tag{32}$$

$$W_{N+1}(b_{N+1}) = \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})], \tag{33}$$

where b_{k+1} is the Bayesian posterior update of b_k given the newly observed $(\boldsymbol{y}_k, \boldsymbol{s}_k)$. The optimal policy is

$$\pi^\star_k(b_k) \;=\; \operatorname{argmax}_{\boldsymbol{s}_k} \; \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})], \tag{34}$$

and the value of the optimal adaptive policy from the prior is

$$V_{\text{adaptive}} \;=\; W_1(p_0). \tag{35}$$

Equations (32)–(33) reduce an infinite-dimensional optimization (over policies = measurable functions) to a sequence of finite-dimensional optimizations (a max over $\boldsymbol{s}_k$ at each step). The price is that W_k is a function on the space of beliefs, which is itself infinite-dimensional; representing W_k exactly is generally impossible except in special cases (linear-Gaussian, finite-state). See Appendix A for the derivation from first principles.

7.3 Oracle value-to-go and the conditional ignorance gap

Analogous to the unconditional ignorance gap, define the *oracle value-to-go* at step k given committed sensors $\boldsymbol{s}_{1:k-1}$:

$$V_k^\star(\boldsymbol{x}) \;=\; \max_{\boldsymbol{s}_{k:N}} \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:k-1}, \boldsymbol{s}_{k:N}). \tag{36}$$

The adaptive policy can never beat the oracle, so $W_k(b_k) \le \mathbb{E}_{\boldsymbol{x}|b_k}[V_k^\star(\boldsymbol{x})]$. The deficit is the **conditional ignorance gap**:

$$\Gamma_k(b_k) \;=\; \mathbb{E}_{\boldsymbol{x}|b_k}[V_k^\star(\boldsymbol{x})] - W_k(b_k) \;\ge\; 0. \tag{37}$$

At $k = 1$ with the prior, $\Gamma_1(p_0) = \mathbb{E}[V_{\text{oracle}}] - V_{\text{adaptive}}$ is the gap that even the *optimal* adaptive policy cannot close. Substituting $W_k = \mathbb{E}[V_k^\star] - \Gamma_k$ into (32) yields a dynamic program on Γ_k—the *gap DP*—which exposes an explicit exploration–exploitation decomposition of the optimal policy. That reformulation is useful for theoretical analysis (bounds on Γ_1, regime classification) but is not required for the remainder of this document; it is standard and available in the optimal-experimental-design literature.

8 Joint Optimization of Geometry and Adaptive Policy

We now combine the two optimizations: over the device geometry $\boldsymbol{c}$ and over the adaptive policy π.

8.1 The joint problem

The full joint problem is

$$\boxed{\max_{\boldsymbol{c}} \; W_1(p_0;\, \boldsymbol{c}),} \tag{38}$$

where $W_1(p_0;\, \boldsymbol{c})$ is the Bellman-optimal value (32) at the prior for geometry $\boldsymbol{c}$, i.e., the joint problem implicitly solves an inner Bellman recursion for each $\boldsymbol{c}$. The geometry enters through every level of the problem: the Jacobians $\boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, the oracle map $\boldsymbol{s}^\star(\boldsymbol{x};\, \boldsymbol{c})$, the transition of beliefs (via the likelihoods), and the terminal $\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$.

8.2 Envelope theorem for the outer gradient

Differentiating (38) naively would require $\partial\pi^\star/\partial\boldsymbol{c}$ (how the optimal policy changes with $\boldsymbol{c}$)—a formidable object. The *envelope theorem* (Danskin) says we don't need it:

$$\boxed{\left.\frac{dW_1}{d\boldsymbol{c}}\right|_{\boldsymbol{c}=\boldsymbol{c}_0} = \left.\frac{\partial}{\partial\boldsymbol{c}}\, \mathbb{E}[\Phi(\boldsymbol{x},\, \pi^\star(\boldsymbol{y};\, \boldsymbol{c}_0),\, \boldsymbol{c})]\right|_{\boldsymbol{c}=\boldsymbol{c}_0},} \tag{39}$$

where $\pi^\star(\cdot;\, \boldsymbol{c}_0)$ is the optimal policy *frozen* at $\boldsymbol{c}_0$, treated as a fixed function of its argument. In words: at the optimal policy for a given geometry, the first-order effect of perturbing the policy vanishes (by the first-order optimality condition on π). We differentiate as if the policy were fixed—only the direct effect of $\boldsymbol{c}$ on the measurement matters, evaluated at whatever sensor values the optimal policy happens to emit.

8.3 Policy iteration scheme

Equation (39) enables an alternating **policy iteration** scheme:

1. **Policy step**: for current $\boldsymbol{c}$, solve or approximate the Bellman recursion (32) to obtain (or approximate) $\pi^\star(\boldsymbol{c})$.

2. **Geometry step**: take a gradient step on $\boldsymbol{c}$ using $\partial\Phi/\partial\boldsymbol{c}$ evaluated at $\pi^\star(\boldsymbol{y};\,\boldsymbol{c})$—the same adjoint/sensitivity computation one would use in the fixed-design problem (24), only now evaluated at the (possibly belief-dependent) sensors produced by the current policy.
3. Repeat until convergence.

Because the Bellman recursion is generically intractable, step 1 is where the computational burden lies; step 2 is cheap by the envelope theorem.

8.4 Reduction by regime

The ignorance gap $\mathbb{E}[IG]$ controls how much of this machinery is actually needed:

Regime	Joint problem reduces to	Solver
$\mathbb{E}[IG] \approx 0$	Fixed D-optimal: $\max_{\boldsymbol{c},\boldsymbol{s}_{1:N}} \mathbb{E}[\Phi]$	Single-level gradient descent
$\mathbb{E}[IG]$ moderate	Certainty-equivalent policy iteration	Oracle map + recursive filter + adjoint
$\mathbb{E}[IG]$ large	Full Bellman DP + policy iteration	Parameterized policy + DP solver

The first column is the problem one *actually* solves; the second is the minimum machinery required. The computability check of Section 5 determines which row applies, *before* any solver is implemented.

Certainty-equivalent (CE) policy. In the moderate-$\mathbb{E}[IG]$ regime, a natural and often near-optimal heuristic is the *certainty-equivalent* policy: at each step, plug the posterior mean $\hat{\boldsymbol{x}}_{k-1} = \mathbb{E}[\boldsymbol{x} \mid b_k]$ into the oracle map and use the resulting sensors,

$$\boldsymbol{s}_k^{\mathrm{CE}} \;=\; \boldsymbol{s}_k^\star(\hat{\boldsymbol{x}}_{k-1}). \tag{40}$$

This replaces the Bellman optimization with a per-step plug-in and is exact as the posterior concentrates; its suboptimality is controlled by the rate of posterior variance reduction and the Lipschitz constant of the oracle map $\boldsymbol{x} \mapsto \boldsymbol{s}^\star(\boldsymbol{x})$.

8.5 The added value of joint optimization

The decoupled program

"*Pick $\boldsymbol{c}$ to maximize the fixed-design objective. Separately, if $\mathbb{E}[IG]$ is large, run an adaptive policy at run time.*"

is *not* equivalent to the joint optimization of (38). The ignorance gap $\mathbb{E}[IG]$ is an upper bound on the adaptive advantage (Remark 4) and is a useful pre-solver diagnostic, but by itself it does not determine either the realized adaptive advantage $V_{\text{adaptive}} - V_{\text{fixed}}$ or the joint-optimum $\boldsymbol{c}^\star$. Three facts together make joint optimization a distinct and necessary step:

1. $\mathbb{E}[IG](\boldsymbol{c})$ and $V_{\text{adaptive}}(\boldsymbol{c}) - V_{\text{fixed}}(\boldsymbol{c})$ are *different functions of* $\boldsymbol{c}$. A $\boldsymbol{c}$ that maximizes the former need not maximize the latter, and vice versa.
2. The ratio $\big(V_{\text{adaptive}} - V_{\text{fixed}}\big)/\mathbb{E}[IG]$—the *EVPI saturation fraction*—varies substantially with $\boldsymbol{c}$, and is typically far below 1 in practice.
3. The $\boldsymbol{c}$ that maximizes V_{adaptive} generally differs from the $\boldsymbol{c}$ that maximizes V_{fixed}: the optimal geometry for an adaptive operator is not the optimal geometry for a pre-committed one.

Radar case study, within-Gaussian-family comparison (see Section 9). For the $K = 16$, $N = 4$ benchmark at $p_{\mathrm{D}} = 0.9$, $p_{\mathrm{FA}} = 0.05$, restricting attention to a single beam family—the Gaussian beam $G_w(\alpha) = \exp(-\alpha^2/(2w^2))$—and comparing the two extremes against the in-family optimum $w^\star = 0.60$ rad gives:

geometry $\boldsymbol{c}$	$\mathbb{E}[IG]$ (EVPI)	V_{adaptive}	realized	saturation
Gaussian $w = 0.05$ (small-w extreme)	2.08 nats	0.62 nats	0.12 nats	5.8%
Gaussian $w = \pi$ (large-w extreme)	0.09 nats	0.12 nats	0.002 nats	2.4%
Gaussian $w = 0.60$ (in-family $w^\star$)	0.64 nats	**0.87** nats	0.11 nats	**17.0**%

The small-w extreme has the *largest* $\mathbb{E}[IG]$ (by a factor of ~ 3 over $w^\star$), yet only the second-largest V_{adaptive} and a saturation of just 5.8%. The large-w extreme has both the smallest $\mathbb{E}[IG]$ and the smallest V_{adaptive}. The in-family optimum $w = 0.60$ *dominates* both extremes on V_{adaptive} (the quantity we actually want to maximize), despite having neither the largest $\mathbb{E}[IG]$ nor the largest realized gain. Routing the geometry choice through EVPI alone would have directed us to the small-w design; the joint DP picks $w = 0.60$. The continuous Gaussian sweep—reported in the companion numerics document—shows interior optima for both V_{adaptive} and V_{fixed}, at *different* values of the beamwidth (0.60 and 0.70 radians respectively), and a realized-gain maximum at yet a third value (0.25 radians). None of these optima would be recovered by EVPI-only or fixed-only geometry selection.

Practical prescription.

1. Compute $\mathbb{E}[IG]$ at a few candidate $\boldsymbol{c}$. If it is uniformly $\ll 1$ nat, adaptation is irrelevant and (24) suffices (Corollary 1).
2. Otherwise, run the joint $(\boldsymbol{c}, \pi)$ optimization (38) via policy iteration with envelope-theorem geometry gradients. The EVPI diagnostic determines *whether* to engage this loop, not *where* its optimum lies.

The separation of concerns is between go/no-go (EVPI) and optimum location (joint DP): they serve complementary but non-interchangeable roles.

9 A Tractable Case Study: Adaptive Radar Beam Search

We conclude with a concrete problem that exercises every ingredient of the framework: the ignorance gap is demonstrably large, the geometry $\boldsymbol{c}$ exerts first-order control on achievable performance, *and* the Bellman recursion (32)–(33) is exactly solvable. The policy-iteration scheme of Section 8 therefore becomes a concrete algorithm rather than an aspirational one. The problem is deliberately small and canonical so that every claim can be verified numerically.

9.1 Problem setup

State. A single target occupies one of K discrete cells at distinct angular positions. We use $K = 16$ cells equally spaced on a circular arc at fixed range from the antenna, so that cell $z \in \{1, \ldots, K\}$ has bearing $\varphi(z) = 2\pi(z-1)/K \in [0, 2\pi)$. The prior p_0 on z is uniform (arbitrary but known priors are equally handled).

Actions. At each step $k = 1, \ldots, N$ with N modest (we use $N = 4$), the operator steers the antenna to a pointing direction $\theta_k \in \Theta = \{\theta_1, \ldots, \theta_M\}$ taken from a finite, symmetric set of $M \sim K$ candidate pointings.

Observations. Each pointing returns a single binary detection $y_k \in \{0,1\}$ with Bernoulli likelihood

$$\Pr(y_k = 1 \mid z, \theta_k;\, \boldsymbol{c}) \;=\; p_{\mathrm{FA}} + (p_{\mathrm{D}} - p_{\mathrm{FA}})\, G_{\boldsymbol{c}}(\theta_k - \varphi(z))\,, \tag{41}$$

where $G_{\boldsymbol{c}}(\alpha) \in [0,1]$ is the normalized antenna gain pattern at angular offset α from the steered main-lobe direction, p_{D} is the on-target detection probability, and p_{FA} is the false-alarm rate.

Geometry. The design variable $\boldsymbol{c}$ is the full vector of antenna-array parameters. Concretely, take the antenna to be a phased array of N_e elements at positions $\{\boldsymbol{r}_n\}_{n=1}^{N_e}$ on the aperture plane, driven by complex excitations $\{a_n\}_{n=1}^{N_e} \in \mathbb{C}^{N_e}$ (each $a_n = |a_n| e^{\mathrm{i}\varphi_n}$ carries both an amplitude *and* a phase). The far-field array factor at angular offset α from the steered direction is

$$F_{\boldsymbol{c}}(\alpha) \;=\; \sum_{n=1}^{N_e} a_n \,\exp\!\big(\mathrm{i}\, k_0\, \boldsymbol{r}_n \cdot \hat{\boldsymbol{\alpha}}\big), \qquad k_0 = 2\pi/\lambda, \tag{42}$$

and the normalized gain pattern used in (41) is $G_{\boldsymbol{c}}(\alpha) = |F_{\boldsymbol{c}}(\alpha)|^2 / \max_{\alpha'} |F_{\boldsymbol{c}}(\alpha')|^2$. The design vector is $\boldsymbol{c} = (\{\boldsymbol{r}_n\}, \{a_n\})$, counted over its real degrees of freedom: $d_c \sim 2N_e$ for a linear array with *fixed* (e.g., half-wavelength-spaced) element positions and only complex excitations as free variables, $\sim 3N_e$ if linear-array positions are also free, and $\sim 4N_e$ for a full planar array (element positions in $\mathbb{R}^2$ plus complex amplitudes). A typical $N_e \in \{8, \ldots, 64\}$ gives d_c in the tens to low hundreds. Receiver parameters $(p_{\mathrm{D}}, p_{\mathrm{FA}})$ are held fixed for clarity, but could be appended to $\boldsymbol{c}$ if they are themselves design variables.

Remark 6 (Why complex excitations matter)**.** *Real non-negative amplitudes (pure amplitude tapers) on a symmetric element layout produce conjugate-symmetric patterns $G_{\boldsymbol{c}}(\alpha) = G_{\boldsymbol{c}}(-\alpha)$, so any null at relative offset $+\alpha$ from the main lobe is mirrored at $-\alpha$. Complex a_n break this mirror symmetry: the extra element-wise phase degree of freedom enables shaped asymmetric main lobes and nulls at* arbitrary *relative offsets (not just symmetric pairs)—a qualitatively richer design space than the real-amplitude subspace. Under standard linear-phase steering the entire pattern still rotates rigidly with the pointing θ_k, so these are offsets relative to the main lobe, not angular positions fixed in absolute terms.*

Terminal reward. Take $\Phi(z, \theta_{1:N}) = \ln K - H(z \mid y_{1:N})$, the information gained about z relative to the uniform prior. Equivalently (for a uniform prior) $\Phi = I(z; y_{1:N})$, the mutual information between the unknown cell and the observation sequence, which is also $\ln\det \boldsymbol{J}_N$ up to an additive constant under the categorical's softmax parametrization. Thus $\Phi \in [0, \ln K]$ with $\Phi = 0$ at the prior and $\Phi = \ln K$ at full resolution. Maximizing Φ concentrates the posterior on the true cell.

9.2 The ignorance gap is large

Oracle. Knowing z, the oracle always points $\theta_k^\star = \varphi(z)$ (or the nearest element of Θ), receiving an on-target Bernoulli with probability p_{D} at every step. After N such measurements, the likelihood ratio for cell z against any competing cell grows as $\big(p_{\mathrm{D}}(1-p_{\mathrm{FA}})/(p_{\mathrm{FA}}(1-p_{\mathrm{D}}))\big)^{\#\text{ detections}}$, which concentrates the posterior on z. For $p_{\mathrm{D}} \gg p_{\mathrm{FA}}$ and N at or above the information-theoretic minimum $\log_2 K$ (at the benchmark $K = 16$, $N = 4$ saturates this), the posterior is essentially fully resolved (entropy $\to 0$), giving $V_{\mathrm{oracle}}(z) \approx \ln K$.

Fixed design (narrow-beam family). Assume the geometry $\boldsymbol{c}$ produces a single-main-lobe, directional gain pattern with main-lobe angular width $\Delta\theta_{\mathrm{ML}}(\boldsymbol{c}) \lesssim 2\pi/K$ (one cell per main lobe) and negligible sidelobes. The fixed design must commit $\theta_{1:N}$ before any data; with $N < K$ narrow beams, at most N distinct cells are illuminated. Conditioning on where the target falls:

- target lies in one of the N probed cells (probability N/K): the on-target beam drives $H \to 0$, so $\Phi = \ln K$;
- target lies in the $K - N$ un-probed cells (probability $(K - N)/K$): all N beams return the off-beam Bernoulli and cannot distinguish the un-probed cells, so $H \approx \ln(K - N)$ and $\Phi \approx \ln K - \ln(K - N) = \ln\big(K/(K - N)\big)$.

Averaging:

$$V_{\text{fixed}} \;\approx\; \ln K \;-\; \frac{K - N}{K} \ln(K - N). \tag{43}$$

Scaling. Combining the two, for designs with narrow single main lobes and budget $N < K$,

$$\boxed{\mathbb{E}[IG] \;=\; V_{\text{oracle}} - V_{\text{fixed}} \;\approx\; \frac{K - N}{K} \ln(K - N) \text{ nats,}} \tag{44}$$

which, for $K \gg N$, scales as $\sim \ln K$; it vanishes continuously as $N \to K$. Note that $\mathbb{E}[IG]$ is always bounded above by the prior entropy $\ln K$, as it must be.

Numerical estimate. Take $K = 16$, $N = 4$, a narrow-beam design with main-lobe width $\Delta\theta_{\text{ML}} \lesssim 2\pi/K = \pi/8$ (e.g., a uniform linear array with $N_e \gtrsim 16$ half-wavelength-spaced elements), $p_{\text{D}} = 0.9$, $p_{\text{FA}} = 0.05$. Equation (44) gives $\mathbb{E}[IG] \approx (12/16) \ln 12 \approx 1.86$ nats, consistent with Monte Carlo evaluation of (28). The EVPI upper bound is substantial—a 2.5× average reduction in the Cramér–Rao uncertainty volume ($e^{\mathbb{E}[IG]/2} \approx e^{0.93}$)—though, as the Rough-sizing discussion below shows, whether adaptation actually *attains* any of that bound is a regime-dependent question, not implied by the size of $\mathbb{E}[IG]$ alone.

9.3 Geometry has first-order impact

The full design $\boldsymbol{c}$ shapes every feature of $G_{\boldsymbol{c}}$ independently: main-lobe width, sidelobe structure, null placement, and grating-lobe behavior. Each of these feeds into the Bernoulli likelihood (41) in a distinct way, and each is independently tunable via the d_c components of $\boldsymbol{c}$:

- *Main-lobe width* $\Delta\theta_{\text{ML}}$ (set primarily by aperture size $L \sim \max_{n,n'} \|\boldsymbol{r}_n - \boldsymbol{r}_{n'}\|$, with $\Delta\theta_{\text{ML}} \sim \lambda/L$ for a well-tapered aperture). Narrow lobes concentrate on-target signal per pointing but can only probe one cell at a time, giving a large $\mathbb{E}[IG]$ in the narrow-beam EVPI sense. Whether adaptation realizes any of that gap is a separate, regime-dependent question analyzed in the Rough-sizing discussion below.
- *First sidelobe level* (controlled by the amplitude taper $\{a_n\}$: uniform taper gives -13 dB, Dolph–Chebyshev or Taylor tapers trade sidelobe level against main-lobe width). Low sidelobes suppress cell-to-cell measurement ambiguity and benefit *both* fixed and adaptive designs.
- *Grating lobes* (set by element spacing: $d > \lambda/2$ introduces copies of the main lobe at other angles). Grating lobes create fundamental ambiguities that no policy can resolve—an aliasing that only $\boldsymbol{c}$ can fix.
- *Null structure at specific relative offsets* (shaped by complex amplitudes and element positions). Deep nulls at designed *relative* offsets from the main lobe allow the measurement to reject specific cell pairs even when the main lobe is steered toward one of them, enhancing cell-pair discrimination in a way that scalar-beamwidth-only parametrizations cannot. Because a phased array steered by linear phase rotates the entire pattern together, these nulls are tied to the main-lobe pointing (their absolute angular positions shift with θ_k); their value is in the pattern's relative structure, not fixed absolute nulls.

Each of these features has first-order impact on both V_{adaptive} and V_{fixed}, and they interact: amplitude tapering that sharpens the main lobe also reduces directivity and raises certain sidelobes. The joint optimum over $(\boldsymbol{c}, \pi)$ is the solution of a d_c-dimensional constrained optimization whose objective is modulated by the adaptive policy that will run on top.

Rough sizing. Binary observations limit the per-step mutual information to at most $\ln 2$ nats, so full resolution of K cells requires at least $N \geq \log_2 K$ measurements regardless of beam design. Under a flat "in-beam / out-of-beam" idealization, write m for the number of cells illuminated per beam; the single-measurement mutual information $I(z; y_k) = H(y_k) - H(y_k \mid z)$ with

$$\Pr(y_k = 1) \;=\; (m/K)\, p_{\mathrm{D}} + ((K-m)/K)\, p_{\mathrm{FA}} \tag{45}$$

is maximized over m at $m^\star$ within $O(1)$ of $K/2$ for any $(p_{\mathrm{D}}, p_{\mathrm{FA}})$ with $p_{\mathrm{D}} > p_{\mathrm{FA}}$. Under this optimum the single-beam main-lobe width is

$$\Delta\theta^\star_{\mathrm{ML}} \;\sim\; \pi \qquad \text{(half the cell ring for } K \text{ cells on a circle).} \tag{46}$$

A key fact about contiguous-arc beams on the ring: each beam has two arc boundaries, so N fixed pointings create at most $2N$ distinct boundary positions and hence at most $2N$ distinguishable segments, each containing $K/(2N)$ cells (in the balanced case). Cells within a segment share the same N-bit label under any *fixed* schedule and cannot be distinguished. Adaptive policies, by contrast, can choose each subsequent pointing so its arc bisects the current posterior support, implementing a binary search that reaches single-cell resolution in $\log_2 K$ steps.

Two regimes follow, at opposite extremes of the beam-width knob:

- *Narrow-beam restriction* ($m \approx 1$, $\Delta\theta_{\mathrm{ML}} \lesssim 2\pi/K$). Each measurement probes one cell. Fixed and adaptive both visit at most N cells in N steps; under the uniform prior they achieve the *same* expected value $V_{\mathrm{fixed}} = V_{\mathrm{adaptive}} \approx \ln K - \frac{K-N}{K}\ln(K-N) \approx 0.91$ nats at $K = 16, N = 4$. The EVPI bound (30) is therefore slack (LHS $= 0$, RHS $= \frac{K-N}{K}\ln(K-N) \approx 1.86$ nats). Large computable $\mathbb{E}[IG]$, zero realized adaptive advantage.
- *Wide-beam regime* ($m \approx K/2$, $\Delta\theta_{\mathrm{ML}} \sim \pi$). Fixed pointings create at most $2N$ segments; at $K = 16, N = 4$ this caps fixed-design resolution at $\ln(2N) = \ln 8 \approx 2.08$ nats. Adaptive binary search reaches full resolution $V_{\mathrm{adaptive}} = \ln K = \ln 16 \approx 2.77$ nats. The adaptive advantage $V_{\mathrm{adaptive}} - V_{\mathrm{fixed}} = \ln(K/(2N)) = \ln 2 \approx 0.69$ nats equals $\mathbb{E}[IG]$ at this $\boldsymbol{c}$, so the EVPI bound is *tight* and adaptation realizes the full gap.

The narrow-beam regime maximizes the *computable* EVPI upper bound but fails to saturate it; the wide-beam regime has a smaller computable $\mathbb{E}[IG]$ but saturates it fully. Joint optimization over $(\boldsymbol{c}, \pi)$ favors the wide-beam design: both $V^\star_{\mathrm{fixed}} = \ln(2N)$ and $V^\star_{\mathrm{adaptive}} = \ln K$ are maximized by $m \approx K/2$. A poor choice of $\boldsymbol{c}$ (sidelobes that confuse specific cell pairs, element spacing introducing grating lobes, or element placement whose $2N$ arc boundaries are badly distributed) wastes the N-step budget. This is exactly the coupling that Section 8's policy iteration is designed to exploit.

9.4 Full Bellman DP is tractable

This is a finite-state, finite-action, finite-observation POMDP with short horizon. Exact solvability follows from three standard facts:

Sondik's theorem (PWLC value functions). For finite POMDPs, each $W_k(b_k)$ is piecewise-linear and convex on the belief simplex. It is represented *exactly* by a finite set of α-vectors $\{\boldsymbol{\alpha}_k^{(i)}\}_{i=1}^{\Gamma_k} \subset \mathbb{R}^K$:

$$W_k(b_k) = \max_i \langle b_k, \boldsymbol{\alpha}_k^{(i)} \rangle. \tag{47}$$

Backward induction from the terminal condition (33)—itself a linear functional of b_{N+1}, hence a single α-vector—produces Γ_{k-1} from Γ_k by a discrete operation (one new vector per $(\theta_k, \boldsymbol{\alpha}_k^{(i_0)}, \boldsymbol{\alpha}_k^{(i_1)})$ triple, modulo dominance pruning).

Dimension count. With $K = 16$, $M = 16$, $|\mathcal{Y}| = 2$, $N = 4$:

- Belief dimension: $K - 1 = 15$.
- Worst-case α-vector growth: $|\Gamma_{k-1}| \leq |\mathcal{A}| \cdot |\Gamma_k|^{|\mathcal{Y}|} = 16 \cdot |\Gamma_k|^2$. Unchecked this is doubly exponential; in practice incremental pruning keeps $|\Gamma_k|$ in the hundreds.
- Memory/time per backup: $O(|\Gamma| \cdot K \cdot M \cdot |\mathcal{Y}|)$—negligible on a laptop.

Off-the-shelf solvers (incremental pruning, SARSOP, PBVI) handle problems of this size in seconds. A hand-coded exact solver in Julia or Python runs in under a minute.

Symmetry reduction. The K-cell ring with a uniform prior and a reflection-symmetric beam pattern carries a dihedral D_K symmetry (cyclic rotations plus one reflection). For $K = 16$ that is $|G| = 32$. The value function inherits the symmetry, $W_k(\sigma \cdot b_k) = W_k(b_k)$ for all $\sigma \in G$, so α-vectors appear in orbits of size up to $|G|$ and only one representative per orbit needs to be explicitly stored. The belief space itself remains $(K-1)$-dimensional—intermediate beliefs generically break the prior's symmetry after the first measurement—but the effective number of distinct α-vectors (and distinct actions, up to symmetry) shrinks by a factor of $|G|$, with corresponding savings in memory and backup time.

In short: step 1 of policy iteration—solve the inner POMDP for the current $\boldsymbol{c}$ (which enters only through the likelihood values $G_{\boldsymbol{c}}(\theta_k - \varphi(z))$ at the $M \cdot K$ action/cell pairs)—is not a research project but a textbook computation. The dependence of the inner DP on $\boldsymbol{c}$ is smooth except at isolated policy-switch boundaries.

9.5 Policy iteration for the joint optimum

With exact inner DP available, the envelope-theorem-based scheme of Section 8 becomes:

1. **Policy step.** For the current design $\boldsymbol{c}_t$, solve the finite POMDP exactly to obtain $\pi^\star(\boldsymbol{c}_t)$ as a depth-N policy tree: every branch corresponds to a value of $y_{1:k-1}$ and maps to an action θ_k.
2. **Geometry step.** For each realized branch of the policy tree, Φ depends on $\boldsymbol{c}$ analytically through the Bernoulli likelihood (41) and the array factor (42) (the derivative of a complex exponential with respect to its amplitude or phase is again a complex exponential). The full vector-valued gradient $\nabla_{\boldsymbol{c}}\Phi \in \mathbb{R}^{d_c}$ is computable in closed form per branch, or by automatic differentiation through the forward model that maps $(\{\boldsymbol{r}_n\}, \{a_n\})$ to the gain values $G_{\boldsymbol{c}}(\theta_k - \varphi(z))$. Aggregate by Monte Carlo over the belief-trajectory distribution induced by $\pi^\star(\boldsymbol{c}_t)$:
$$\nabla_{\boldsymbol{c}} W_1\big|_{\boldsymbol{c}=\boldsymbol{c}_t} \;=\; \mathbb{E}_{z,\, y_{1:N}\sim\pi^\star(\boldsymbol{c}_t)}[\nabla_{\boldsymbol{c}}\Phi(z,\, \pi^\star(y_{1:N};\boldsymbol{c}_t),\, \boldsymbol{c})]_{\boldsymbol{c}=\boldsymbol{c}_t}\,. \tag{48}$$
Because $\pi^\star(\boldsymbol{c}_t)$ is held fixed by the envelope theorem, the right-hand side requires no policy Jacobian $d\pi^\star/d\boldsymbol{c}$—a quantity that is ill-defined at policy-switch boundaries where the argmax in (34) changes.
3. **Gradient step.** Update $\boldsymbol{c}_{t+1} = \mathrm{Proj}_{\mathcal{C}}(\boldsymbol{c}_t + \eta\, \nabla_{\boldsymbol{c}} W_1)$ and repeat, where $\mathcal{C}$ is the feasible set (aperture bounds, per-element power constraints $|a_n| \leq a_{\max}$ or total-power constraint $\sum_n |a_n|^2 \leq P$, minimum-element-spacing to control mutual coupling, etc.) and $\mathrm{Proj}_{\mathcal{C}}$ is a suitable projection. Each complex a_n is parametrized by its real pair $(a_n^{\mathrm{re}}, a_n^{\mathrm{im}}) \in \mathbb{R}^2$, and the gradient is taken with respect to these real components; no Wirtinger calculus is required, and the whole optimization runs in real parameter space $\mathbb{R}^{d_c}$.

Because the inner DP in step 1 is solved *exactly*, there is no approximation error in the gradient direction (48)—only Monte Carlo variance (over z and the stochastic $y_{1:N}$ trajectory), shrinking as $|\text{samples}|^{-1/2}$. Convergence of the outer loop is governed by the smoothness of $W_1(\cdot)$ as a function of $\boldsymbol{c}$, which is continuous except on the codimension-one boundaries where $\pi^\star$ becomes non-unique; the envelope theorem remains valid through these, with a subgradient taking the place of the gradient.

9.6 What this problem illuminates

The radar case study stresses every moving part of the framework at minimum dimension:

- $\mathbb{E}[IG]$ **as a go/no-go gate.** Setting $N = K$ (budget equals number of cells) allows the fixed design to visit every cell, collapsing $\mathbb{E}[IG] \to 0$—the framework correctly predicts that adaptation becomes pointless and the aperture-size optimum degenerates (any aperture large enough to resolve cells suffices). The phase transition at $N = K$ is sharp and numerically verifiable.
- **The DP diagnoses the regime.** The DP produces a different $\pi^\star$ in each of the two beam-width regimes, and the EVPI bound (30) reacts accordingly. In the narrow-beam regime ($m \approx 1$) the DP returns a policy that ties any fixed cell-by-cell schedule, correctly reporting zero adaptive benefit despite a large computable $\mathbb{E}[IG] \approx 1.86$ nats—the EVPI is slack, and the case study shows how to detect a slack EVPI numerically. In the wide-beam regime ($m \approx K/2$) the DP returns a binary-search policy that strictly beats any fixed schedule by the full $\mathbb{E}[IG] = \ln(K/(2N)) \approx 0.69$ nats, saturating the EVPI. Either way the DP is *right about which regime it is in*, and the explore–exploit structure in the wide-beam solution emerges without any hand-crafted policy parametrization, making the case study a discriminating test of the whole framework.
- **Geometry–policy coupling is quantitatively predicted.** The binary-partition main-lobe scaling (46) can be tested numerically against the policy-iteration fixed point; agreement confirms that the envelope-theorem gradient is correctly capturing the joint optimum along the main-lobe-width dimension of $\boldsymbol{c}$. The *remaining* dimensions of $\boldsymbol{c}$—amplitude taper, sidelobe structure, null placement, element positions—are shaped by components of the gradient (48) that no scalar scaling argument can predict, providing a genuine multi-dimensional test of the framework.

Adjacent problems. The same template covers several other canonical testbeds amenable to exact DP:

- *Sequential material identification by spectroscopy*: finite set of candidate chemicals as hypotheses, actions = excitation wavelengths from a discrete grid, observations = quantized absorbance; geometry $\boldsymbol{c}$ = spectrometer linewidth and wavelength grid.
- *Adaptive compressed sensing with known sparsity*: state = support set of $s \ll d$ active indices out of d ambient dimensions (discrete state space of cardinality $\binom{d}{s}$), actions = dictionary rows, observations = quantized linear projections or thresholded detections (finite alphabet); geometry $\boldsymbol{c}$ = dictionary design. For $(d, s) = (10, 2)$ there are 45 states; with observations quantized to a finite alphabet the resulting POMDP remains exactly tractable.
- *Medical-imaging sequential slicing*: discrete lesion-location hypotheses, actions = scan parameters, observations = quantized intensities; geometry $\boldsymbol{c}$ = field-of-view and slice-selection gradients.

All three share the property that finite state/action/observation spaces plus short horizon keep the Bellman recursion exactly solvable, while nonlinear measurement likelihoods keep $\mathbb{E}[IG] > 0$ and the geometry variable meaningful.

10 When the joint DP is intractable: a hierarchy of principled relaxations

The radar case study of Section 9 was engineered to make the exact joint DP tractable: finite state space ($K = 16$), finite observation alphabet ($\{0, 1\}$), finite pointing grid (K actions), and short horizon ($N = 4$). Many sensor-design problems of practical interest—continuous target state, continuous action space, expensive forward models, high-dimensional geometry $\boldsymbol{c}$—violate

at least one of these assumptions. When the exact Bellman recursion is beyond reach, the framework of Sections 7–8 remains the correct organizing picture, but it must be instantiated with approximations of the inner optimization. This section catalogs the principled relaxations, ordered by how much structure they preserve, so that the subsequent photonic case study can be positioned unambiguously within this taxonomy.

10.1 Sources of intractability

Three structural barriers recur:

1. **Belief-space size.** For a K-point discrete state, the reachable belief tree from a uniform prior has up to 2^N leaves after N binary observations—manageable for $K = 16, N = 4$, infeasible for $K = 10^6$ or continuous $\boldsymbol{x}$. For continuous $\boldsymbol{x}$ the belief is an infinite-dimensional density; one must either discretize, project onto a parametric family (e.g., Gaussian), or carry a particle representation.
2. **Action-space size.** When $\boldsymbol{s}_k$ parametrizes a continuous object (beamwidth, amplitude taper, phase profile), the inner $\max_{\boldsymbol{s}_k}$ inside each Bellman backup is itself a nonconvex continuous optimization, not an enumerable finite argmax.
3. **Per-step cost.** A single evaluation of $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ or $V_{\text{oracle}}(\boldsymbol{x})$ in physics-based sensor models typically requires one or more forward electromagnetic solves (FDFD, FDTD, RCWA), turning each Bellman node into a minutes-long computation. Even a modestly sized belief tree becomes intractable.

Any one of these collapses the exact DP; in the photonic setting all three bite simultaneously.

10.2 Relaxation path 1: neural adaptive policies (deep RL)

Parametrize the adaptive policy by a neural network $\pi_\theta : \mathcal{B} \to \mathcal{S}$ mapping beliefs (or observation histories) to actions, and train θ jointly with $\boldsymbol{c}$ by policy gradient, REINFORCE, differentiable-simulator pathwise gradients, or any combination thereof. The full joint objective becomes

$$\max_{\boldsymbol{c},\,\theta} \; \mathbb{E}_{\boldsymbol{x}\sim p_0,\, \boldsymbol{y}_{1:N}\sim p(\cdot|\boldsymbol{x},\boldsymbol{c},\pi_\theta)} \big[\, \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}^{\pi_\theta}, \boldsymbol{c}) \,\big], \tag{49}$$

with $\boldsymbol{s}_k^{\pi_\theta} = \pi_\theta(b_k)$ and b_k the online belief.

What this preserves. Multi-step reasoning (the NN has access to the full history), arbitrary action geometry, and—if the forward model is differentiable in $\boldsymbol{c}$—a direct pathwise gradient $\partial\Phi/\partial\boldsymbol{c}$. Modern end-to-end optical learning (Sitzmann et al., 2018; Chang & Wetzstein, 2019) fits this template with π_θ collapsed to a one-step reconstruction network.

What this gives up. Optimality certificates: π_θ is a function approximator, not a Bellman solver, and the envelope theorem of Section 8.2 applies cleanly only at an interior optimum of θ. In practice one carries both gradients $\partial/\partial\boldsymbol{c}$ and $\partial/\partial\theta$, subject to RL's standard sample-complexity and training-stability pathologies.

A characteristic failure mode: hardware-light fixed points. Joint training of $(\boldsymbol{c}, \theta)$ inherits a pathology familiar from end-to-end optical learning: the NN policy π_θ can absorb the training signal and *compensate* for a poor $\boldsymbol{c}$ by investing in policy complexity, leaving $\boldsymbol{c}$ near its initialization. The mechanism is gradient asymmetry. $\partial\Phi/\partial\theta$ propagates through cheap NN backprop and is high-rank; $\partial\Phi/\partial\boldsymbol{c}$ propagates through an expensive forward-model adjoint (FDFD or FDTD) and is lower-rank per step. The net effect is that θ converges to a local optimum of the *generic* $\Phi(\cdot; \boldsymbol{c}_0)$ evaluated at the *initialization* $\boldsymbol{c}_0$, and the optimization settles into a "policy-heavy, hardware-light" fixed point where $\boldsymbol{c}$ is near-trivial and the NN does all the work. In regimes where geometry has first-order impact on V_{adaptive}—a property Section 9

quantifies with a 2.8× separation between the best and worst within-family geometries on the radar benchmark—this failure mode silently returns a strictly worse answer than the bi-level paths (2a)–(2c) below.

Standard countermeasures (none fully eliminates the risk): (i) capacity control on π_θ (bottleneck width, weight decay, dropout); (ii) two-timescale updates with $\boldsymbol{c}$ stepped at a larger rate than θ, or alternating schedules that freeze one while training the other; (iii) warm-starting θ from a prescribed info-theoretic rule (path (2b) or (2c) below) so that the first gradient signal reaching $\boldsymbol{c}$ is not obscured by early NN chaos; (iv) explicit regularizers penalizing "easy" $\boldsymbol{c}$ (e.g., a penalty on $\|\boldsymbol{c} - \boldsymbol{c}_0\|$ below some threshold, or on high NN entropy that indicates the policy is doing too much disambiguation work). In our experience the most reliable of these is (iii): a geometry that has been shaped by a principled inner rule is a much better starting point for deep-RL refinement than the reverse.

10.3 Relaxation path 2: bi-level optimization with a *prescribed* inner policy

A cheaper and more interpretable relaxation prescribes the policy π in closed form as an information-theoretic *rule*, then optimizes $\boldsymbol{c}$ by the outer loop of Section 8 (envelope-theorem gradient + policy iteration) with the prescribed rule standing in for the Bellman-optimal policy. The outer-level problem becomes

$$\max_{\boldsymbol{c}} \ \mathbb{E}\big[\, \Phi(\boldsymbol{x},\, \boldsymbol{s}_{1:N}[b.;\boldsymbol{c}],\, \boldsymbol{c}) \,\big] \qquad \text{subject to the chosen rule } \boldsymbol{s}_k = \mathsf{R}_k(b_k;\boldsymbol{c}), \tag{50}$$

and the outer gradient propagates to $\boldsymbol{c}$ through the *implicit* dependence of the inner argmax on $\boldsymbol{c}$, by the implicit function theorem (IFT)—an exact computation even when the resulting $\boldsymbol{s}_k$ is not globally optimal in the Bellman sense.

The choice of rule R_k determines how much of the original Bellman structure survives.

(2a) Oracle certainty-equivalent control

At step k, form a point estimate $\hat{\boldsymbol{x}}_k = \operatorname{argmax}_{\boldsymbol{x}} b_k(\boldsymbol{x})$ or $\hat{\boldsymbol{x}}_k = \mathbb{E}[\boldsymbol{x} \mid b_k]$, then play the first step of the schedule the clairvoyant oracle would have chosen against $\hat{\boldsymbol{x}}_k$:

$$\boldsymbol{s}_k^{\mathrm{CE}} = \big(\boldsymbol{s}^\star(\hat{\boldsymbol{x}}_k;\boldsymbol{c})\big)_1, \qquad \boldsymbol{s}^\star(\boldsymbol{x};\boldsymbol{c}) = \operatorname{argmax}_{\boldsymbol{s}_{1:N-k+1}} \Phi(\boldsymbol{x},\, \boldsymbol{s}_{1:N-k+1},\, \boldsymbol{c}). \tag{51}$$

This is *receding-horizon* and therefore non-myopic, but it assumes $\hat{\boldsymbol{x}}_k$ is correct and plays the oracle designed for that specific $\boldsymbol{x}$. Under narrow-beam geometries $\boldsymbol{s}^\star(\boldsymbol{x})$ degenerates to a single pointing (point-and-verify), inducing a textbook exploration deficit: if $\hat{\boldsymbol{x}}_k$ is wrong, the policy may deterministically never test the true target. Computationally, (51) still requires an oracle solve per step; if V_{oracle} was the bottleneck, this relaxation replaces one expensive call with N.

(2b) Myopic Bayesian D-optimal (one-step mutual-information maximization)

Choose each $\boldsymbol{s}_k$ to maximize the expected posterior entropy reduction from the *next* observation only:

$$\boldsymbol{s}_k^{\text{myopic}} \ = \ \operatorname{argmax}_{\boldsymbol{s}_k} \ I(\boldsymbol{x}; y_k \mid b_k, \boldsymbol{s}_k, \boldsymbol{c}) \ = \ \operatorname{argmax}_{\boldsymbol{s}_k} \ \ln\det \mathbb{E}_{\boldsymbol{x}\sim b_k}\big[\boldsymbol{J}(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})\big]. \tag{52}$$

No look-ahead beyond one step; no multi-step value bootstrap. Greedy information gain is $(1 - 1/e)$-competitive against the clairvoyant DP optimum for submodular objectives (Krause & Guestrin, 2008; Golovin & Krause, 2011), making this the default baseline in sequential design. Two variants must be distinguished: $\ln\det \mathbb{E}[\boldsymbol{J}]$ (Bayesian Fisher) versus $\mathbb{E}[\ln\det \boldsymbol{J}]$ (expected log-det); they coincide only when $\boldsymbol{J}$ is state-independent. For tractability the first is preferred; for fidelity to the posterior volume, the second.

(2c) Gaussian–trace surrogate with recursive inference

A further stack of approximations—each justified by a specific operating regime—yields the cheapest member of the family:

$$\boldsymbol{s}_k^{\mathrm{BFIM}} \;=\; \mathrm{argmax}_{\boldsymbol{s}_k}\ \mathrm{tr}\,\boldsymbol{J}\big(\bar{\boldsymbol{x}}_k,\, \boldsymbol{s}_k,\, \boldsymbol{c}\big), \qquad \bar{\boldsymbol{x}}_k = \mathbb{E}[\boldsymbol{x} \mid b_k], \tag{53}$$

coupled with an *extended Kalman filter* (EKF) for the recursive belief update. The three stacked simplifications, relative to (2b), are:

(i) tr in place of ln det (A-optimal instead of D-optimal). tr is cheaper (linear in diagonal entries, no factorization) and admits an analytic adjoint for $\boldsymbol{c}$-gradients; it loses scale-invariance and weights all parameter directions equally.

(ii) Evaluate $\boldsymbol{J}$ at the posterior *mean* $\bar{\boldsymbol{x}}_k$ instead of integrating $\mathbb{E}_{\boldsymbol{x}\sim b_k}[\boldsymbol{J}(\boldsymbol{x},\cdot)]$. Exact when $\boldsymbol{J}$ is constant in $\boldsymbol{x}$, first-order accurate when the posterior is narrower than the scale on which $\boldsymbol{J}(\boldsymbol{x})$ varies. Analogous to a Laplace approximation.

(iii) EKF for b_k—a Gaussian posterior $\mathcal{N}(\bar{\boldsymbol{x}}_k, \boldsymbol{\Sigma}_k)$ propagated by linearized mean and covariance updates. Exact in the linear–Gaussian limit; robust when the posterior remains unimodal.

Each layer individually is a standard move. Stacked, they reduce the inner per-step solve from max-of-expectation-of-log-det (path 2b) to an analytic max-of-trace at a single $\boldsymbol{x}$-point, and the outer geometry gradient $\partial/\partial\boldsymbol{c}$ flows through (53) by IFT in closed form. The result is an end-to-end bi-level pipeline whose cost is dominated by the outer forward-model solves, not by the inner policy step.

10.4 The hierarchy and what each entry buys

Table 1: Tractability hierarchy for adaptive sensor design. Each row buys the column "cost" by giving up something in "loses." The exact DP is the gold standard the radar case study of Section 9 exhibits; the deep-RL and bi-level relaxations are the routes when that gold standard is out of reach.

	inner step per $\boldsymbol{s}_k$	look-ahead	posterior handling	guarantee
Exact Bellman DP	full belief backup	N-step	exact	joint-optimal
Deep RL (§10.2)	one NN forward pass	$\leq N$, implicit	NN-internal	approximate, black-box
Oracle CE (2a)	oracle at $\hat{\boldsymbol{x}}_k$	$(N-k+1)$-step CE	point estimate	none; under-explores
Myopic Bayes-D (2b)	argmax $\ln\det \mathbb{E}[\boldsymbol{J}]$	1-step	full posterior	$(1-1/e)$ via sub-modularity
Gaussian–trace (2c)	argmax $\mathrm{tr}\,\boldsymbol{J}(\bar{\boldsymbol{x}}_k)$	1-step	Gaussian / EKF	none; empirically sharp in near-Gaussian regimes

Two observations about this ladder. First, *every row* is a bi-level problem in $\boldsymbol{c}$: the inner step is a function of the current belief and of $\boldsymbol{c}$, and the geometry gradient flows through it by IFT. The "joint optimization of geometry and policy" developed in Section 8 is structurally the same in all five rows; what changes is only the fidelity of the inner optimization. Second, rows 3–5 (the prescribed-rule family) are not mutually exclusive with row 2 (deep RL): a deep-RL policy can be *initialized* from a prescribed information-theoretic rule, amortizing the rule's structure across a distribution of problem instances—a hybrid design that, in our experience, combines the sample efficiency of (2b)–(2c) with the flexibility of (1).

10.5 Transition: the photonic sensor problem as an instance of (2c)

The remainder of this work develops, in full detail, one worked instance of path (2c) applied to a 2D photonic sensor: a cross-waveguide device whose permittivity distribution is the geometry variable $\boldsymbol{c}$, whose sensor parameters $\boldsymbol{s} = (\phi_1, \phi_2)$ are input-port phases, and whose observations $\boldsymbol{y}$ are port powers. Each of the three bulleted simplifications (i)–(iii) above is explicitly justified in that context:

- The state $\boldsymbol{x}$ is a small refractive-index perturbation (dimensionless range $\Delta n \in [10^{-5}, 10^{-4}]$), so the map $\boldsymbol{y} = f(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) + \boldsymbol{\eta}$ is well-approximated by a Taylor expansion of f in $\boldsymbol{x}$, making the EKF posterior a sharp approximation to the true posterior. This justifies (iii).
- The Fisher information $\boldsymbol{J}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ varies slowly in $\boldsymbol{x}$ over the posterior support under standard laser-noise statistics, so evaluation at $\bar{\boldsymbol{x}}_k$ in place of $\mathbb{E}[\boldsymbol{J}(\boldsymbol{x})]$ carries negligible error. This justifies (ii).
- The $\operatorname{tr} \boldsymbol{J}$ surrogate, while not invariant under parameter scaling, agrees closely with $\ln \det \boldsymbol{J}$ for the small parameter dimensions of this problem ($\dim \boldsymbol{x} \leq O(10^2)$), and admits an analytically differentiable inner argmax with a closed-form IFT adjoint. This justifies (i).
- The full outer geometry objective $\mathbb{E}[\|\boldsymbol{\mu}_N - \boldsymbol{x}_0\|^2]$ is differentiable in $\boldsymbol{c}$ end-to-end, via a custom FDFD adjoint ($\partial \boldsymbol{S}/\partial \varepsilon$), the BFIM inner rule (53) with its IFT-based derivative, and the EKF recursive updates with their own manual matrix adjoints. This realizes the full Section 8 envelope-theorem outer loop in an operating regime where the exact Bellman DP is multiple orders of magnitude beyond laptop reach.

The photonic problem is therefore not an alternative *framework*; it is an instantiation of the joint–$(\boldsymbol{c}, \pi)$ framework of this paper at row (2c) of the hierarchy in Table 1, with each approximation justified by the physics of the operating regime. The radar case study of Section 9 and the photonic case study that follows are two points on the same methodological spectrum: one where the gold-standard DP is tractable and used as such, one where the gold-standard DP is unattainable and is replaced by the cheapest principled relaxation. The message of both is the same: *joint* optimization of $(\boldsymbol{c}, \pi)$ is the correct object; the difference is only in which inner solver is affordable.

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A Derivation of Bellman's Equation

We derive equations (32)–(33) from first principles. The argument is general to any finite-horizon sequential decision problem; we specialize the notation to the setting of this document.

A.1 Problem setup

Recall the ingredients:

- An unknown state $\boldsymbol{x} \sim p_0$ (the prior).
- Horizon N with steps $k = 1, 2, \ldots, N$.
- At step k, an action $\boldsymbol{s}_k \in \mathbb{R}^{d_s}$ is chosen, then $\boldsymbol{y}_k = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k$ is observed, $\boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$.
- An adaptive policy $\pi = (\pi_1, \ldots, \pi_N)$ with $\pi_k : \boldsymbol{y}_{1:k-1} \mapsto \boldsymbol{s}_k$.
- A terminal reward $\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})$ accrued only at the end.

The goal is to find

$$V_{\text{adaptive}} \;=\; \max_{\pi} \; \mathbb{E}_{\boldsymbol{x},\, \boldsymbol{y}_{1:N}}[\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}))]\,, \tag{54}$$

a maximization over functions—hopeless at face value.

A.2 Sufficient statistic: the belief state

Because $\boldsymbol{x}$ is the only unknown and the likelihoods $p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k)$ are known, **all information in $\boldsymbol{y}_{1:k-1}$ relevant to future decisions is carried by the posterior belief**

$$b_k(\boldsymbol{x}) \;=\; p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1}). \tag{55}$$

Two observation histories yielding the same b_k are indistinguishable going forward, so we may search over policies $\pi_k : b_k \mapsto \boldsymbol{s}_k$ without loss. Boundary cases: $b_1 = p_0$ (prior, no data yet), b_{N+1} is the posterior after all N observations.

The belief updates by Bayes' rule: given $(b_k, \boldsymbol{s}_k, \boldsymbol{y}_k)$,

$$b_{k+1}(\boldsymbol{x}) \;=\; \frac{p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k)\, b_k(\boldsymbol{x})}{\int p(\boldsymbol{y}_k \mid \boldsymbol{x}', \boldsymbol{s}_k)\, b_k(\boldsymbol{x}')\, d\boldsymbol{x}'}. \tag{56}$$

The marginal distribution of $\boldsymbol{y}_k$ used for expectations in the Bellman equation is

$$p(\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k) \;=\; \int p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k)\, b_k(\boldsymbol{x})\, d\boldsymbol{x}. \tag{57}$$

A.3 Value-to-go: definition

Define the *value-to-go* at step k given belief b_k and committed sensors $\boldsymbol{s}_{1:k-1}$ (the latter suppressed in the notation):

$$W_k(b_k) \;=\; \max_{\pi_k, \pi_{k+1}, \ldots, \pi_N} \; \mathbb{E}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}) \mid b_k, \pi_{k:N}]\,. \tag{58}$$

Boundary cases:

- $k = 1$: $W_1(p_0) = V_{\text{adaptive}}$—the quantity we want.
- $k = N + 1$: no decisions remain; $\boldsymbol{s}_{1:N}$ are all committed; the terminal value is simply the expected reward under the final belief,

$$W_{N+1}(b_{N+1}) \;=\; \mathbb{E}_{\boldsymbol{x} \mid b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})]\,. \tag{59}$$

A.4 Peeling off one step

Start from (58) and split the outer max into "choose $\boldsymbol{s}_k$ now" and "choose the remaining $\pi_{k+1:N}$":

$$W_k(b_k) \;=\; \max_{\boldsymbol{s}_k} \max_{\pi_{k+1},\ldots,\pi_N} \mathbb{E}[\Phi \mid b_k, \boldsymbol{s}_k, \pi_{k+1:N}]\,. \tag{60}$$

Apply the *tower property* (law of total expectation) by conditioning on $\boldsymbol{y}_k$:

$$\mathbb{E}[\Phi \mid b_k, \boldsymbol{s}_k, \pi_{k+1:N}] = \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[\mathbb{E}[\Phi \mid b_{k+1}, \pi_{k+1:N}]]\,, \tag{61}$$

using the fact that b_{k+1} is a deterministic function of $(b_k, \boldsymbol{s}_k, \boldsymbol{y}_k)$ and all remaining randomness depends on b_{k+1} only through the belief (Markov property of beliefs).

Substituting into (60) and pushing $\max_{\pi_{k+1:N}}$ *inside* $\mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}$ (justified below):

$$W_k(b_k) \;=\; \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}\left[\max_{\pi_{k+1},\ldots,\pi_N} \mathbb{E}[\Phi \mid b_{k+1}, \pi_{k+1:N}]\right]. \tag{62}$$

The inner bracket is exactly $W_{k+1}(b_{k+1})$ by the definition (58) at step $k+1$. Therefore:

$$\boxed{W_k(b_k) \;=\; \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,.} \tag{63}$$

A.5 Why the push-in of the max is legal

In general $\mathbb{E}[\max_i X_i] \geq \max_i \mathbb{E}[X_i]$ (Jensen, since max is convex)—the inequality is strict, not an equality. So why is the move in (63) legal?

Because we are *not* exchanging max with $\mathbb{E}$ abstractly. We are exploiting that $\pi_{k+1:N}$ may *depend on* $\boldsymbol{y}_k$ (through b_{k+1}). A policy that observes $\boldsymbol{y}_k$ before committing the continuation is strictly more powerful than one that commits the continuation in advance—and it is the former that problem (54) allows. Formally, for each realization of $\boldsymbol{y}_k$ we choose the optimal continuation pointwise, and maximizing pointwise inside the expectation gives the same value as maximizing over all measurable continuation plans outside. This is the *interchangeability lemma* for adaptive optimization—the rigorous form of Bellman's *principle of optimality*: an optimal policy's continuation is optimal from whatever state results from its first decision.

A.6 The full Bellman system

Combining the recursion and the boundary condition:

$$\begin{aligned}
W_{N+1}(b_{N+1}) \;&=\; \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})]\,, && (64)\\
W_k(b_k) \;&=\; \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,, \quad k = N, N-1, \ldots, 1, && (65)\\
\pi_k^\star(b_k) \;&=\; \operatorname{argmax}_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,, && (66)\\
V_{\text{adaptive}} \;&=\; W_1(p_0). && (67)
\end{aligned}$$

A.7 Action-value (Q) decomposition

Define the action-value

$$\mathcal{Q}_k(b_k, \boldsymbol{s}_k) \;=\; \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,. \tag{68}$$

Then $W_k(b_k) = \max_{\boldsymbol{s}_k} \mathcal{Q}_k(b_k, \boldsymbol{s}_k)$ and $\pi_k^\star = \operatorname{argmax}_{\boldsymbol{s}_k} \mathcal{Q}_k$. This is the standard action-value formulation used in reinforcement learning and approximate dynamic programming.

A.8 Sanity check at the terminal step

At $k = N$, using the terminal (64):

$$W_N(b_N) = \max_{\boldsymbol{s}_N} \mathbb{E}_{\boldsymbol{y}_N|b_N,\boldsymbol{s}_N}\left[\mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})]\right] \tag{69}$$

$$= \max_{\boldsymbol{s}_N} \mathbb{E}_{\boldsymbol{x}|b_N}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N-1}, \boldsymbol{s}_N)]\,, \tag{70}$$

where the second line uses the tower property (for any integrable f not depending on $\boldsymbol{y}_N$, $\mathbb{E}_{\boldsymbol{y}_N|b_N,\boldsymbol{s}_N}\mathbb{E}_{\boldsymbol{x}|b_{N+1}}[f(\boldsymbol{x})] = \mathbb{E}_{\boldsymbol{x}|b_N}[f(\boldsymbol{x})]$). The last step commits no future action, so the optimal $\boldsymbol{s}_N$ is simply the certainty-equivalent choice under the current belief—consistent with the intuition that explore/exploit balance collapses to pure exploit at the final step.

A.9 Why the equation is powerful and cursed

Powerful. The Bellman equation reduces an intractable optimization over *functions* (policies on observation histories) to a recursion over *single-step* decisions. If one can evaluate the max and $\mathbb{E}$ at each step (e.g., by discretizing actions and observations, or by parametric approximation), the problem is solved by N sweeps of dynamic programming.

Cursed. The belief b_k is a probability distribution over $\boldsymbol{x}$—a point in an infinite-dimensional space—and $W_k(\cdot)$ is a function on this space. Representing W_k exactly is generally impossible except in special cases: (i) *linear-Gaussian*, where b_k is fully parametrized by its mean and covariance and W_k admits a closed form via Kalman-filter-based DP; (ii) *finite discrete states*, where b_k is a vector on the simplex and W_k is piecewise-linear and convex (Sondik). For general nonlinear problems, approximations are unavoidable: parametric W_k, particle-filter beliefs, certainty-equivalent policies, or the Jensen-gap bounds of the gap DP.

A.10 Summary

Starting from

$$V_{\text{adaptive}} = \max_{\pi} \mathbb{E}[\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}))]\,, \tag{71}$$

three moves yield the Bellman equation:

(a) **Sufficient statistic reduction**: replace history $\boldsymbol{y}_{1:k-1}$ with belief b_k.

(b) **Peel off one action**: $\max_{\pi_{k:N}} = \max_{\boldsymbol{s}_k} \max_{\pi_{k+1:N}}$.

(c) **Tower property with interchangeability**: push $\max_{\pi_{k+1:N}}$ inside $\mathbb{E}_{\boldsymbol{y}_k}$, revealing $W_{k+1}(b_{k+1})$.

The recursion

$$W_k(b_k) = \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,, \qquad W_{N+1}(b_{N+1}) = \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi] \tag{72}$$

rests on exactly two facts: **the Markov property of beliefs under Bayesian updating**, and **the principle of optimality** (the optimal continuation can depend on intervening observations without loss).